

ADMINISTRATION HVAC - CENTRAL PAU DUCT AND INSTRUMENTATION SCHEMATIC

OUTDOOR / ROOF PLANT

RETURN MAIN
3006.68 L/s ASSUMED

PDI = LOCAL GAUGE; PDT / DPT = PLC PRESSURE INPUT
Each unit: OEM refrigerant HP / LP / oil safety, compressor delay and motor protections.
Catalogue coil duty, available external total pressure, weights and access clearances: TBC.

INDOOR COMMON SUPPLY / RETURN SECTIONS

ROOF MAIN BASIS SA-M01 1200x650; 3697.80 L/s; 9 ROOM DROPS (D-022)

SOURCE: HAP / ADMIN ARCHITECTURE / C-H-SS-001 Rev0. All new equipment and instrument tags are proposed.

GENERAL NOTES / DESIGN HOLDS

- All duct sizes are clear internal W x H in mm; round sizes are diameters. Add approved insulation outside these dimensions.
- HAP reference supply 3697.80 L/s. Assumed return 3006.68 L/s. Final IAQ, outdoor air, revised thermal loads and OEM fan pressure remain open.
- Room targets: +50 Pa normal rooms; +25 Pa kitchen, toilet and clean-agent relative to outdoors. Actual pressure / outward paths are unverified.
- Confirmed fire stops the entire HVAC system. Fire ESD restart requires local manual reset; post-fire purge is a separate approved design.
- Reference substation drawings guide drafting style only. Project requirements follow the supplied TAQA specification and accepted room basis.
- 3 x 50% PAUs: two duty, one available. Any unit may be standby; programmed rotation and fault changeover apply.
- Rev11 roof option: common heater / conditional humidifier in proposed weather-protected roof enclosure; room comfort heaters indoors. Roof suitability / final duty TBC.

FOR COORDINATION - NOT FOR CONSTRUCTION

Unconfirmed routes, levels, outdoor air and OEM selections: Insufficient data to verify.

CLIENT / DESIGN BASIS

TAQA TRANSMISSION

N-19082-RO-012 / Design Standardization for Pumping Stations

DRAWING TITLE

Administration HVAC - central PAU duct and instrumentation schematic

DRAWING NUMBER

ADM-HVAC-S-001

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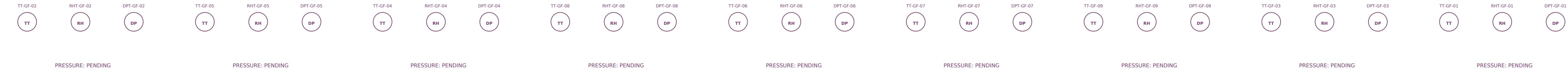
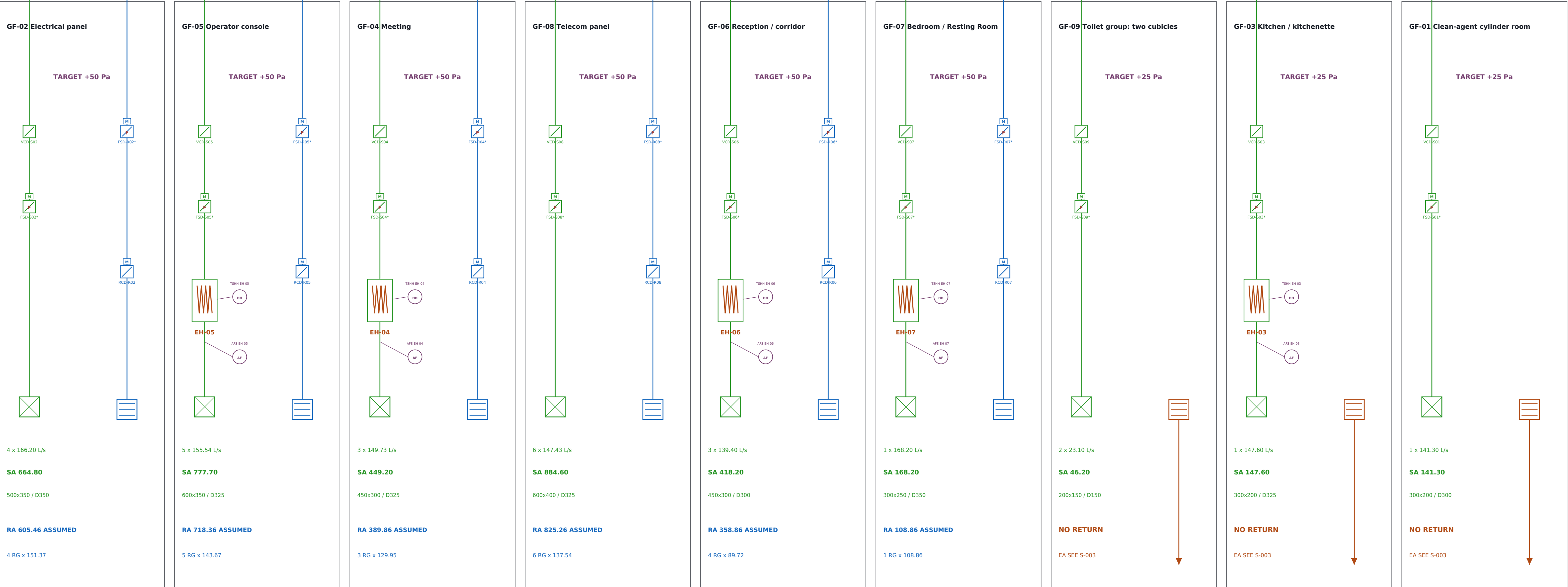
SCALE NTS | A1 / mm / L/s / Pa

Prepared: draft / Checked / approved: pending

ADMINISTRATION HVAC - ROOM SUPPLY AND RETURN SCHEMATIC

PAU COMMON SUPPLY: 3697.80 L/s

DUCTED RETURN: 3006.68 L/s ASSUMED; CAG / KITCHEN / TOILET EXCLUDED



DAMPERS / TERMINALS

Supply diffusers and return grilles: integral OBD, tagged neck, face, flow, plenum, test point and access. Drawn room branch paths have VCD; return pressure-control damper RCD is additional. Main and each room temperature / humidity / pressure values report to HPCP-01. Pressure sensor reference is outdoors, not the adjacent room.

ROOM AIRFLOW BALANCE

For the six +50 Pa return rooms, supply - return = 59.34 L/s as a NET outward allowance. It includes the net effect of transfer, relief and envelope outflow; it is not an additional HAP duct loss. Allocate actual paths before pressure verification. +25 Pa extract rooms must meet both minimum ACH and the full room / transfer balance.

SA = cooled mixed PAU air. Delivered room outdoor air must be checked separately from total supply / extract ACH.

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- Room targets: +50 Pa normal rooms; +25 Pa kitchen, toilet and clean-agent relative to outdoors. Actual pressure / outward paths are unverified.
- Confirmed fire stops the entire HVAC system. Fire ESD restart requires local manual reset; post-fire purge is a separate approved design.
- Reference substation drawings guide drafting style only. Project requirements follow the supplied TAQA specification and accepted room basis.
- The common net equivalent area is 0.01 m2 per +50 Pa return room. C=0.65, density=1.2 and delta-P=50 give 59.34 L/s offset per room.
- RCD return modulation is a proposed pressure-control provision. Confirm range, fail state and tuning; fixed flows are initial balancing values.
- FSD shown with * is conditional on approved fire compartmentation. Exact ratings, locations and fire positions remain TBC.

FOR COORDINATION - NOT FOR CONSTRUCTION

Unconfirmed routes, levels, outdoor air and OEM selections: Insufficient data to verify.

CLIENT / DESIGN BASIS

TAQA TRANSMISSION

N-19082-RO-012 / Design Standardization for Pumping Stations

DRAWING TITLE

Administration HVAC - room supply and return schematic

DRAWING NUMBER

ADM-HVAC-S-002

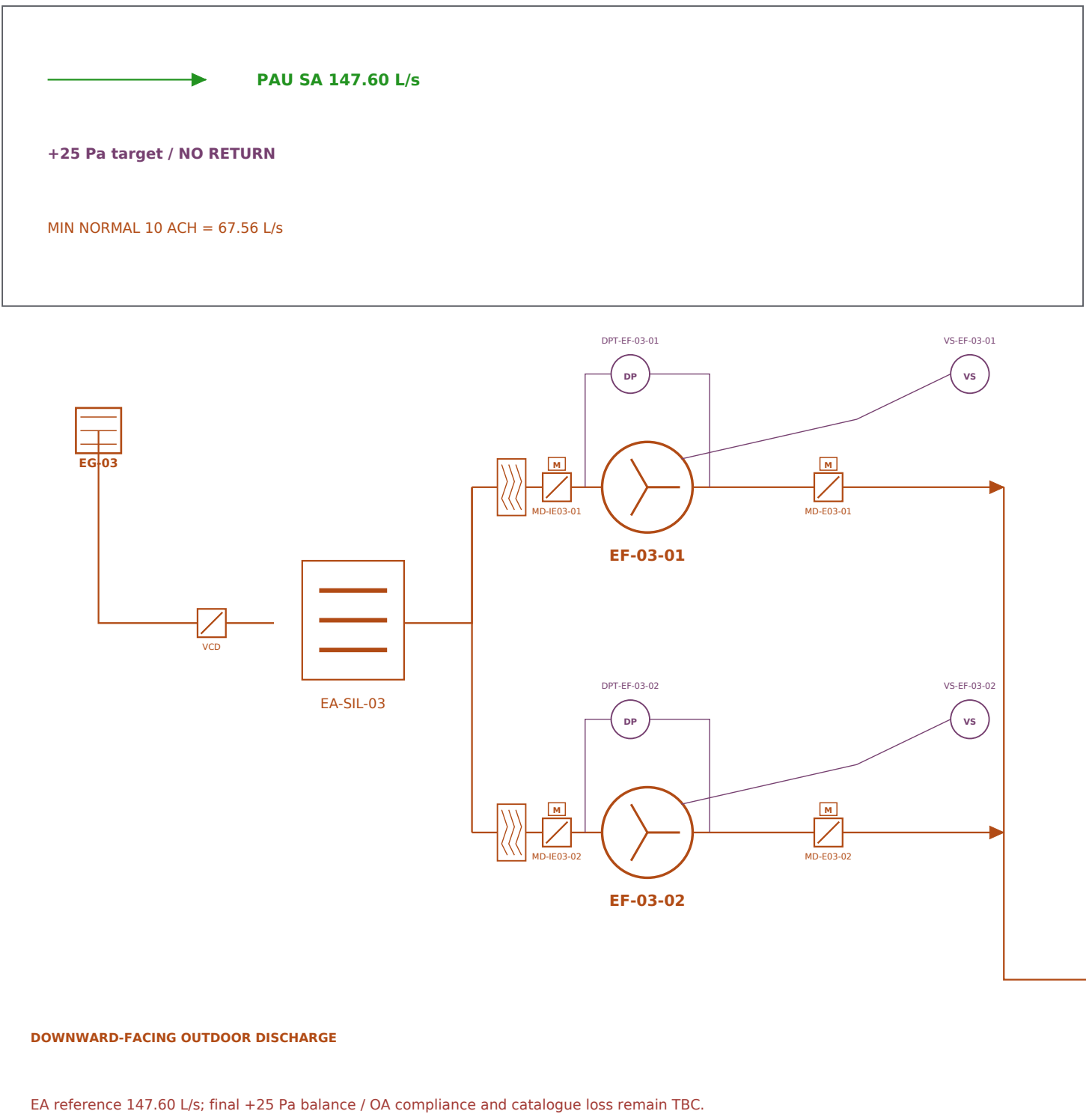
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SCALE NTS | A1 / mm / L/s / Pa

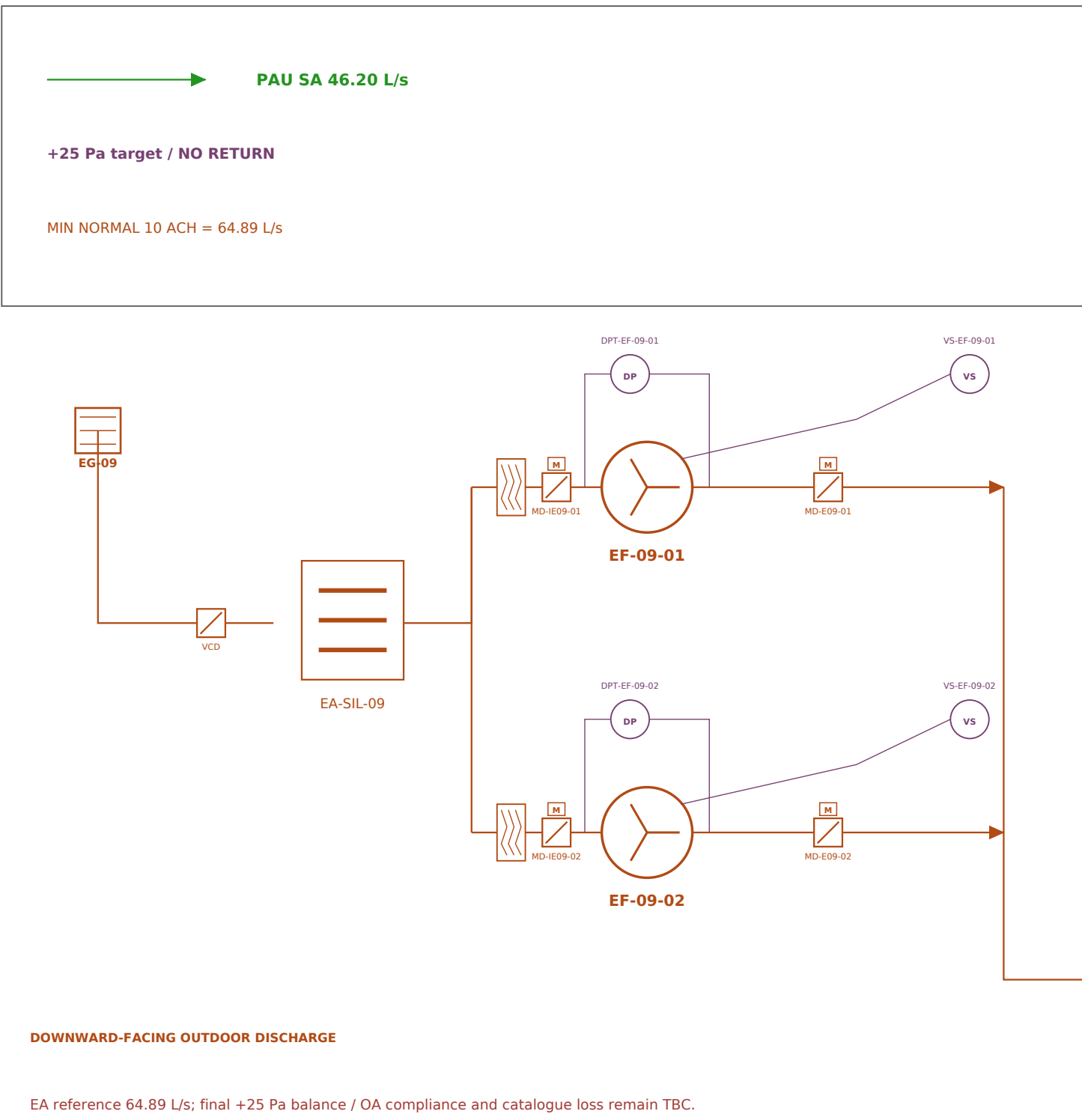
Prepared: draft / Checked / approved: pending

ADMINISTRATION HVAC - DEDICATED EXTRACTS AND CABLE BASEMENT

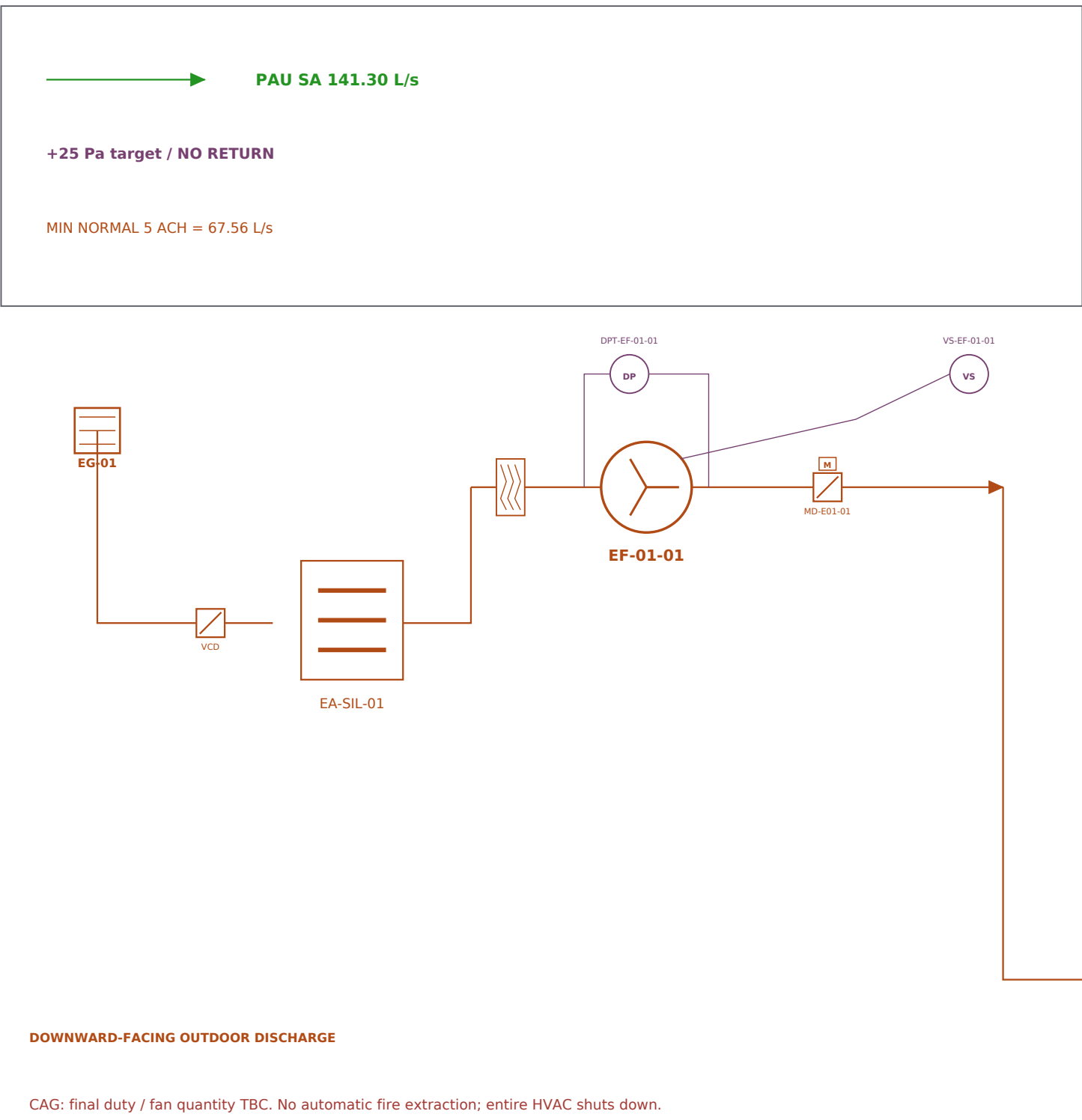
GF-03 KITCHEN / KITCHENETTE



GF-09 TOILET GROUP: TWO CUBICLES

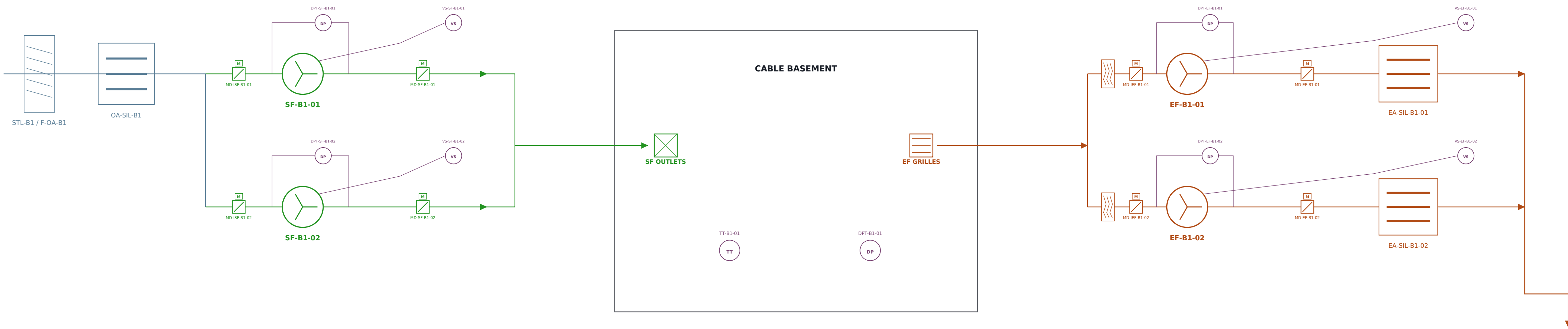


GF-01 CLEAN-AGENT CYLINDER ROOM



CABLE BASEMENT - SEPARATE VENTILATION CIRCUIT

AIRFLOW DUTY TBC: $Q \geq \max(4 \times V / 3.6, 1000 \times \text{heat_kW} / (1.06 \times 1.0216 \times 5))$ L/s



NORMAL / FIRE / POST-FIRE

Normal fans satisfy temperature, fresh-air, ACH and pressure criteria together. Confirmed fire stops ALL HVAC fans and heaters, including CAG and basement. Fire damper positions follow the approved compartment and cause-and-effect design. Same-fan post-fire purge remains a separate proposal; no smoke-control fan rating or automatic purge is established here.

Basement plan, duct sizes, equipment clearances, penetrations, fan curve and full pressure loss circuit: TBC.

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- Room targets: +50 Pa normal rooms; +25 Pa kitchen, toilet and clean-agent relative to outdoors. Actual pressure / outward paths are unverified.
- Confirmed fire stops the entire HVAC system. Fire ESD restart requires local manual reset; post-fire purge is a separate approved design.
- Reference substation drawings guide drafting style only. Project requirements follow the supplied TAQA specification and accepted room basis.
- Kitchen and toilet: 2 x 100% extract with automatic duty rotation. No central return from these rooms.
- Clean-agent: normal 5 ACH minimum; dedicated extract, fan quantity / pickup level / agent suitability TBC.
- Basement: separate 2 x 100% supply + 2 x 100% extract; greater of 4 ACH and heat removal. Volume and heat load are missing.

FOR COORDINATION - NOT FOR CONSTRUCTION

Unconfirmed routes, levels, outdoor air and OEM selections: Insufficient data to verify.

CLIENT / DESIGN BASIS

TAQA TRANSMISSION

N-19082-RO-012 / Design Standardization for Pumping Stations

DRAWING TITLE

Administration HVAC - dedicated extracts and cable basement

DRAWING NUMBER

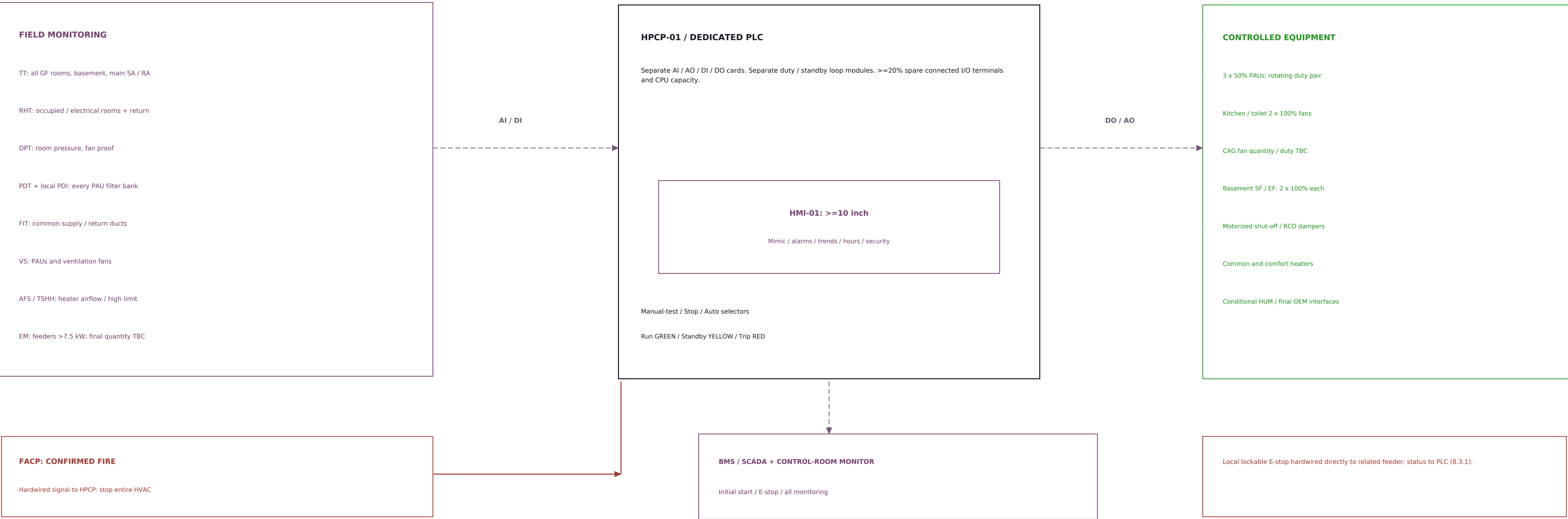
ADM-HVAC-S-003

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SCALE NTS | A1 / mm / L/s / Pa

Prepared: draft Checked / approved: pending

ADMINISTRATION HVAC - CONTROLS, I/O AND CAUSE-AND-EFFECT



PRELIMINARY WIRED I/O ALLOCATION

Type	Shown points	20% spare minimum	Minimum channels
AI	65	13	78
AO	22	5	27
DI	141	29	170
DO	44	9	53

Full unique tag register, equipment list and individual points are included as CSV files in the CAD package. COMM values and OEM-internal safety circuits are not counted as wired AI/DI points. All new tags are draft allocations.

DRAFT OPERATING SEQUENCE / CAUSE-AND-EFFECT

Trigger	Input	Equipment action	Damper / permissive action	Restart / limits
Confirmed fire	FACP -> HPCP	STOP all PAUs, extract and basement fans; heaters off	Damper fire positions per approved cause-and-effect; not inferred	Local manual reset after fire clears
Normal start	HMI / BMS start + Auto	Prove duty path dampers; start chosen duty pair and duty extract fans	Standby PAU and parallel fan inlet / outlet paths isolated	Sequence / delays by approved OEM logic
Duty PAU fault	Unit fault / fan DP	Alarm; isolate failed unit; enable available alternate	Maintain two healthy 50% units when available	Automatic fault changeover; minimum-runtime delay TBC
Low / high room pressure	Room DPT	Proposed RCD / OA or extract-speed adjustment within airflow constraints	Respect cooling, fresh-air and minimum ACH limits	Alarm if targets cannot be met; revise supply / balance
Extract duty fan fault	Fan fault / proof	Start standby for kitchen / toilet / basement	Prove available path; isolate failed path	CAG standby arrangement TBC
Heater high limit / no flow	TSHH / AFS	Heater OFF	Fans / dampers per OEM sequence; fire overrides	Manual or OEM reset only after fault clears
Power loss / return	Mains / feeder status	HVAC stops; normal restart on restored power	Fire ESD latch overrides automatic restart	Automatic power-return restart only without fire latch
Post-fire purge	Separate approved command	Not enabled by this drawing	Agent, fire fan suitability and makeup to be approved	No automatic purge assumed

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- Confirmed fire stops the entire HVAC system. Fire ESD restart requires local manual reset; post-fire purge is a separate approved design.
- Reference substation drawings guide drafting style only. Project requirements follow the supplied TAQA specification and accepted room basis.
- Control logic, alarm limits, delays, communication protocol and actuator fail positions require I&C / OEM approval.
- Room DPTs beyond the TAQA minimum and extract fan pressure proof are explicit added provisions, not quoted minimum requirements.
- Point schedule is preliminary. CAG quantity, conditional fire dampers, humidifier and final energy-meter feeder count can change I/O.

FOR COORDINATION - NOT FOR CONSTRUCTION

Unconfirmed routes, levels, outdoor air and OEM selections: Insufficient data to verify.

CLIENT / DESIGN BASIS

TAQA TRANSMISSION

N-19082-RO-012 / Design Standardization for Pumping Stations

DRAWING TITLE

Administration HVAC - controls, I/O and cause-and-effect

DRAWING NUMBER

ADM-HVAC-S-004

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SCALE NTS | A1 / mm / L/s / Pa

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